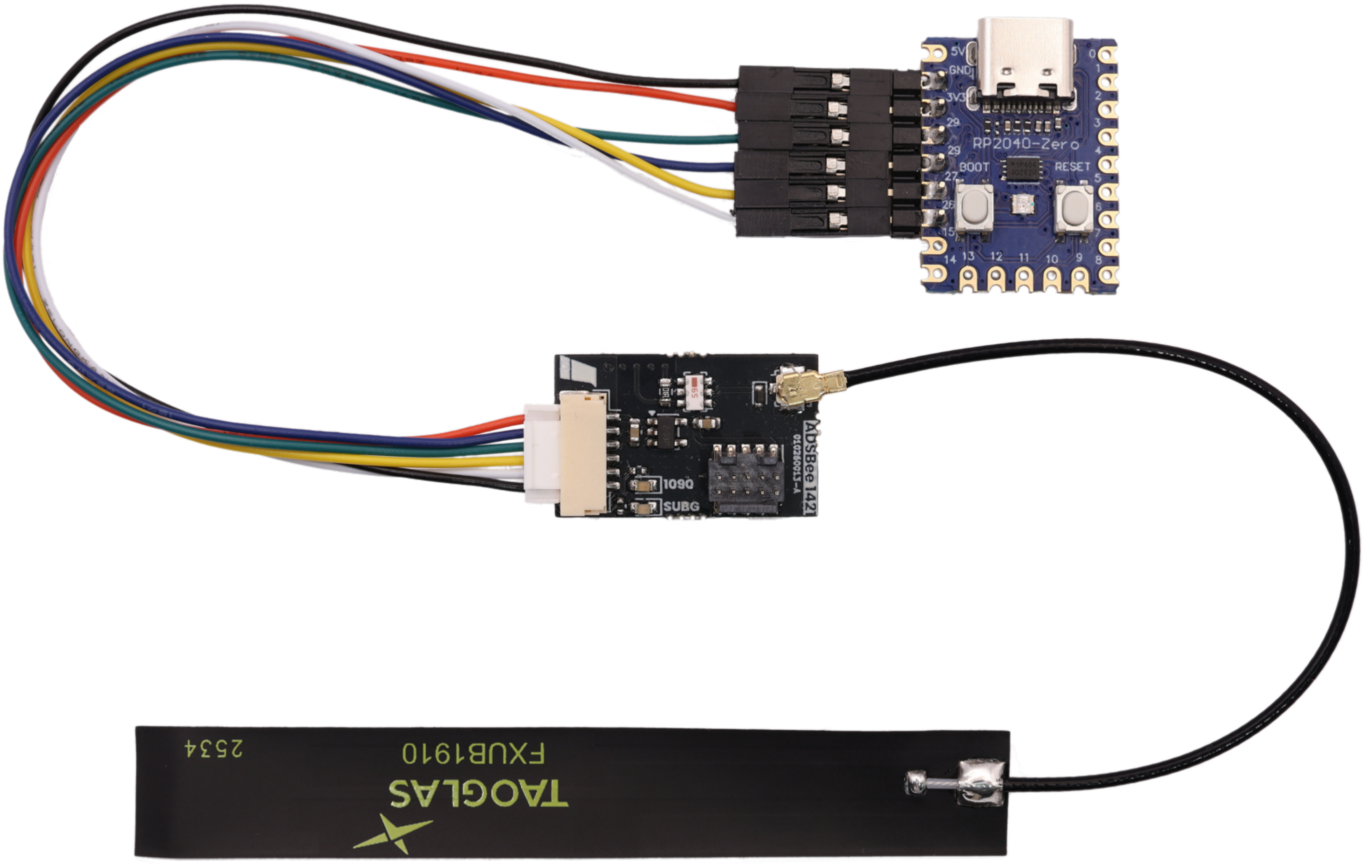
ADSBee 1421 Devkit

Ultra Low-Power Dual Band ADS-B Receiver Module
Programming + Evaluation Kit



Contents

Revision History

20260806

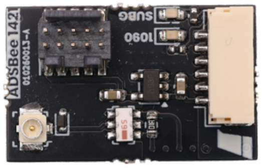
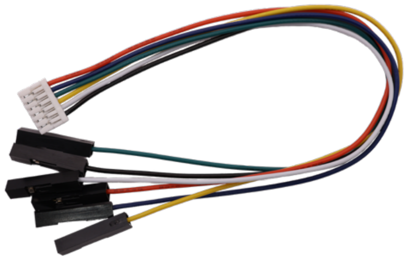
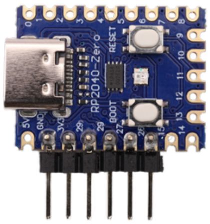
- Initial release.

1 Overview

This kit contains everything you need to get started developing with ADSBee m1421! The included PCBA is a variant of the ADSBee 1421 with an included 10-pin ARM SWD programming header, allowing breakpoint debugging on the ADSBee m1421 module's CC1314 RF microcontroller using a compatible debug probe (Segger J-Link or equivalent probe that supports cJTAG is recommended).

The kit also includes a wire harness that breaks out the ADSBee 1421's 6-pin JST-GH header to 0.1" DuPont pin headers, allowing it to be connected to arbitrary interface devices, or the included ADSBee 1421 Programmer board.

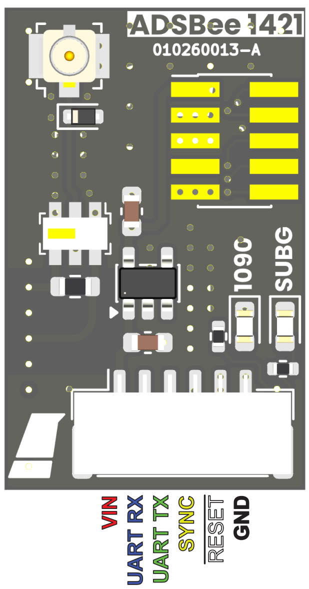
2 Kit Contents

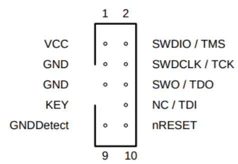
Part Image	PN	Qty	Description
ADSBee 1421 Devkit PCBA 	010260017	1	Dual-band ADS-B receiver devkit. Same as the ADSBee 1421, but includes a 10-pin ARM SWD header to allow breakpoint debugging via cJTAG.
Taoglas FXUB1910 1090/978MHz Dual-band Flex Antenna 	070260005	1	Dual-band ADS-B flex antenna.
ADSBee 1421 JST-GH to DuPont Pin Socket Breakout 	050260006	1	Cable for connecting ADSBee 1421 to programmer.
ADSBee 1421 Programmer PCBA 	010260018	1	ADSBee 1421 programmer and serial bridge.

3 Device Pinouts

3.1 ADSBee 1421 Devkit PCBA

ADSBee 1421 Connector Pinouts





10-PIN ARM SWD CONNECTOR

All lines are TTL logic levels (0-VCC).

Connects to CC1314R10 via cJTAG or 4-wire JTAG.

Includes VCC sense and GND.

COMMS CONNECTOR

All lines are TTL logic levels (0-VCC).

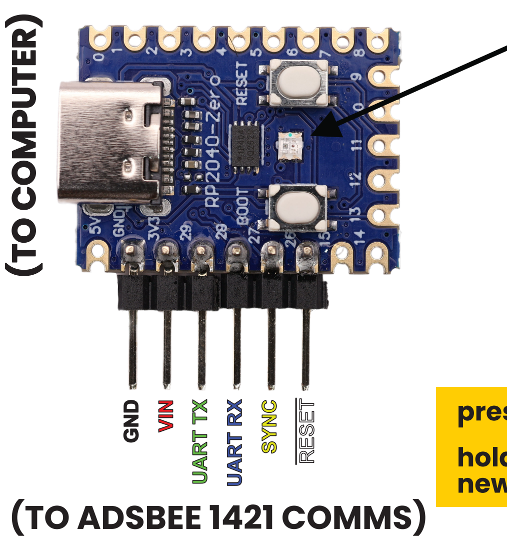
VIN range is 1.8-5.5VDC.

Hold SYNC high on RESET rising edge to enter UART bootloader.

UART interface is "CONSOLE" and can be used for control commands, reporting protocols, and logging.

3.2 ADSBee Programmer

ADSBee 1421 Programmer



LED Color

- yellow blink: waiting for device
- orange: forced reflash armed
- cyan: CRC check before flash
- magenta blink: flashing
- blue blink: post-flash CRC verify
- blue: console baud negotiation
- green: console pass-through
- red: error

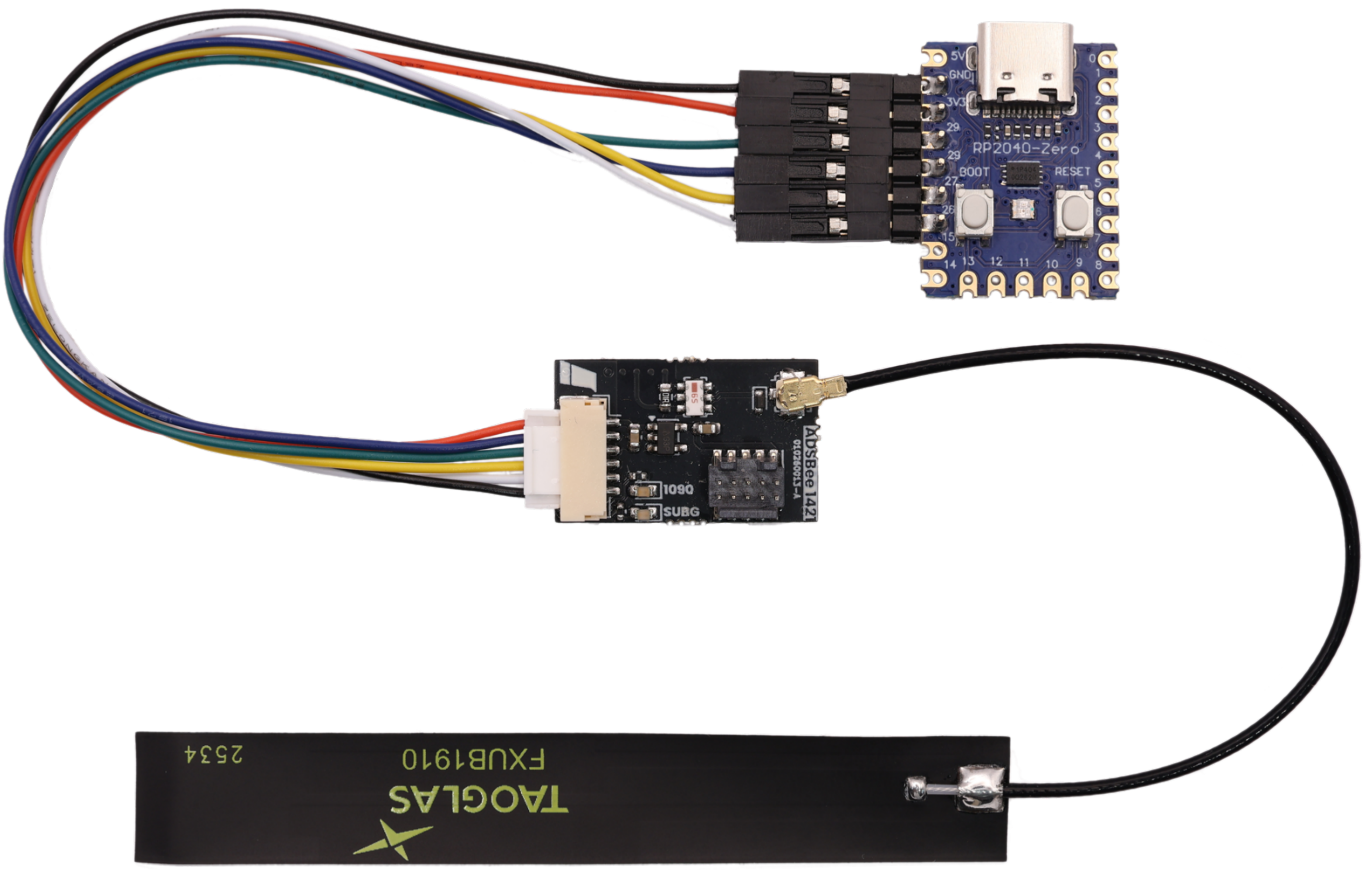
press BOOT to start flashing a 1421

hold BOOT and press RESET to load new firmware image from PC

4 Evaluation

4.1 Connection Diagram

For programming and evaluation, connect ADSBee 1421 to the ADSBee 1421 programmer as shown below.

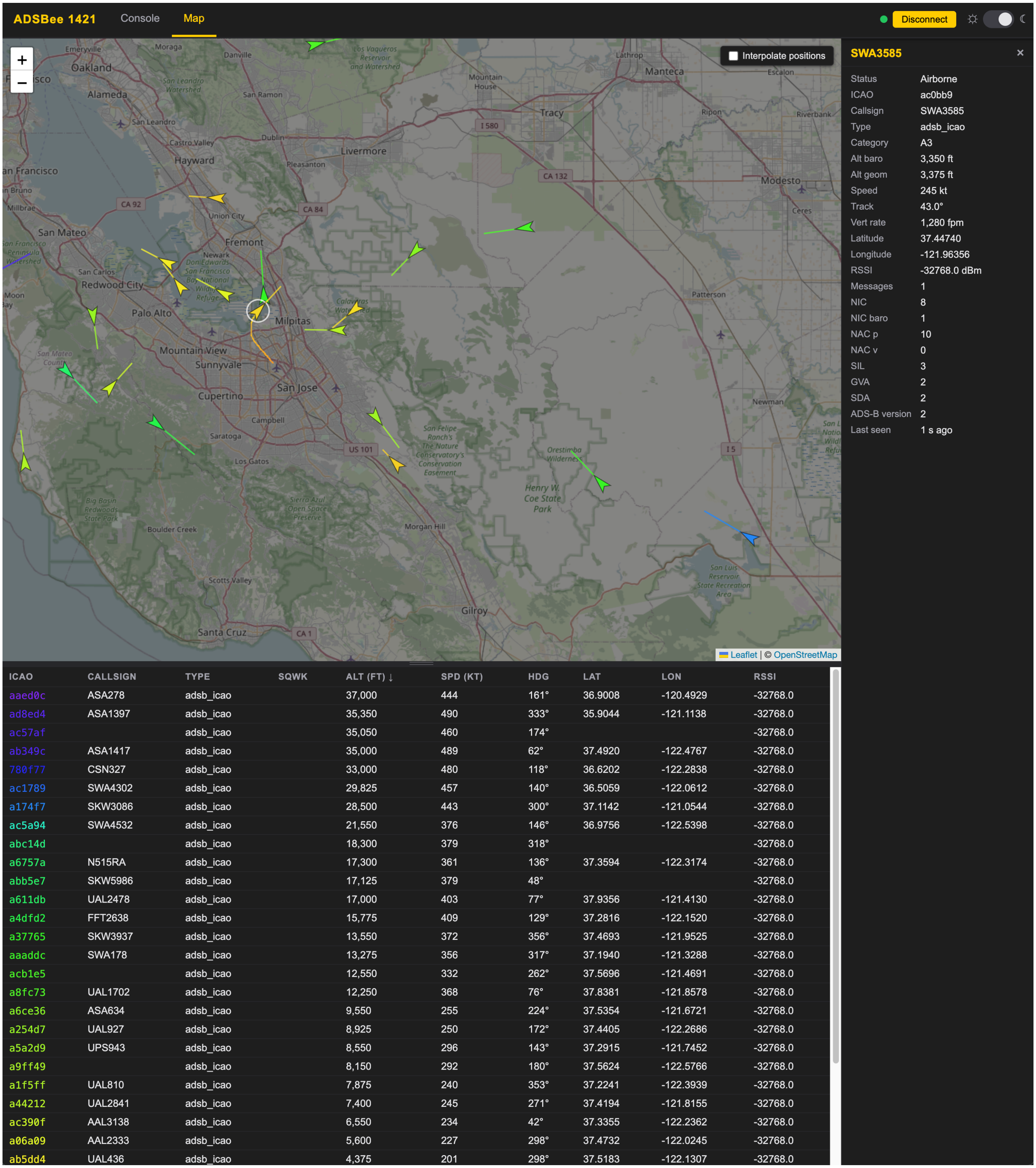


The ADSBee 1421 Programmer will enumerate as a USB CDC device called “ADSBee”. Once the light on the ADSBee 1421 Programmer turns green, it functions as a serial bridge and connects the ADSBee 1421 to the computer. A standard UART to USB converter can be substituted if desired.

Note that the default baud rate on the ADSBee 1421 module is 1000000 baud. Changing the baud rate from this value could result in the module being unable to communicate via the ADSBee 1421 Programmer.

4.2 Aircraft Map

A live aircraft map is available from adsbee.aero/1421-console. Use the yellow “Connect” button on the top right corner of the webpage to connect to your ADSBee 1421 via web serial, then click on the “Map” tab to see aircraft you are receiving. The HTML page may also be downloaded for local use (it’s just a single HTML file)!



ICAO	CALLSIGN	TYPE	SQWK	ALT (FT) ↓	SPD (KT)	HDG	LAT	LON	RSSI
aaed0c	ASA278	adsb_icao		37,000	444	161°	36.9008	-120.4929	-32768.0
ad8ed4	ASA1397	adsb_icao		35,350	490	333°	35.9044	-121.1138	-32768.0
ac57af		adsb_icao		35,050	460	174°			-32768.0
ab349c	ASA1417	adsb_icao		35,000	489	62°	37.4920	-122.4767	-32768.0
780f77	CSN327	adsb_icao		33,000	480	118°	36.6202	-122.2838	-32768.0
ac1789	SWA4302	adsb_icao		29,825	457	140°	36.5059	-122.0612	-32768.0
a174f7	SKW3086	adsb_icao		28,500	443	300°	37.1142	-121.0544	-32768.0
ac5a94	SWA4532	adsb_icao		21,550	376	146°	36.9756	-122.5398	-32768.0
abc14d		adsb_icao		18,300	379	318°			-32768.0
a6757a	N515RA	adsb_icao		17,300	361	136°	37.3594	-122.3174	-32768.0
abb5e7	SKW5986	adsb_icao		17,125	379	48°			-32768.0
a611db	UAL2478	adsb_icao		17,000	403	77°	37.9356	-121.4130	-32768.0
a4dfd2	FFT2638	adsb_icao		15,775	409	129°	37.2816	-122.1520	-32768.0
a37765	SKW3937	adsb_icao		13,550	372	356°	37.4693	-121.9525	-32768.0
aaaddc	SWA178	adsb_icao		13,275	356	317°	37.1940	-121.3288	-32768.0
acb1e5		adsb_icao		12,550	332	262°	37.5696	-121.4691	-32768.0
a8fc73	UAL1702	adsb_icao		12,250	368	76°	37.8381	-121.8578	-32768.0
a6ce36	ASA634	adsb_icao		9,550	255	224°	37.5354	-121.6721	-32768.0
a254d7	UAL927	adsb_icao		8,925	250	172°	37.4405	-122.2686	-32768.0
a5a2d9	UPS943	adsb_icao		8,550	296	143°	37.2915	-121.7452	-32768.0
a9ff49		adsb_icao		8,150	292	180°	37.5624	-122.5766	-32768.0
a1f5ff	UAL810	adsb_icao		7,875	240	353°	37.2241	-122.3939	-32768.0
a44212	UAL2841	adsb_icao		7,400	245	271°	37.4194	-121.8155	-32768.0
ac390f	AAL3138	adsb_icao		6,550	234	42°	37.3355	-122.2362	-32768.0
a06a09	AAL2333	adsb_icao		5,600	227	298°	37.4732	-122.0245	-32768.0
ab5dd4	UAL436	adsb_icao		4,375	201	298°	37.5183	-122.1307	-32768.0

5 Updating Firmware

The included ADSBee 1421 Programmer board began its life as a humble RP2040 Zero module, but with the addition of some pin headers and firmware, has been transformed into a serial bridge and firmware updater compatible with any ADSBee 1421 / m1421 device.

5.1 Updating an ADSBee 1421 / m1421

Each ADSBee 1421 Programmer includes a ADSBee 1421 firmware built into its flash, and when connected to an ADSBee 1421 / m1421 device over UART (with SYNC and RESET also connected), it will automatically connect to the attached ADSBee and bring its firmware up to date. Once the firmware update is complete, the ADSBee 1421 Programmer will function as a USB to serial converter, allowing the user to interact with the ADSBee 1421 / m1421 via USB.

5.2 Updating an ADSBee Programmer

To update the firmware image baked into an ADSBee 1421 Programmer, follow the steps below.

1) Download the latest firmware release from the releases page in the ADSBee github repository (https://github.com/CoolNamesAllTaken/adsbee/releases). The firmware file will be labelled "adsbee_1421_programmer-X.X.X.uf2", where X.X.X is the firmware version of the baked-in firmware image (e.g. 0.3.6), and uf2 is the filetype.
2) Connect the ADSBee 1421 Programmer to your computer via USB. Hold the BOOT button and click RESET; the ADSBee 1421 Programmer will show up as a flash drive labelled RPI-RP2. Let go of both buttons.
3) Drag and drop the adsbee_1421_programmer-X.X.X.uf2 firmware file onto the RPI-RP2 drive and allow the transfer t complete. The ADSBee 1421 Programmer now has a new firmware and is ready to update ADSBee 1421 / m1421 devices via UART.